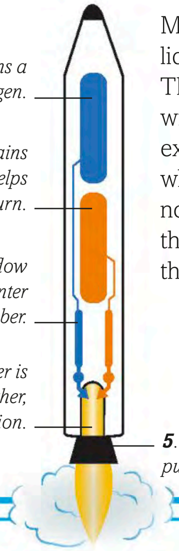
Uses

Hydrogen-filled balloon

This balloon *can rise high into the atmosphere where sensors gather information about atmospheric pressure, temperature, and wind speed.*

HOW ROCKET FUEL WORKS



***1**. This chamber contains a fuel called liquid hydrogen.*

***2**. This chamber contains liquid oxygen, which helps the hydrogen burn.*

***3**. Pumps control the flow of the liquids as they enter the combustion chamber.*

***4**. The combustion chamber is where the liquids mix together, creating an explosion.*

***5**. The nozzle emits hot vapour, pushing the rocket upwards.*

Many space rockets use liquid hydrogen as a fuel. The hydrogen reacts with oxygen to form extremely hot steam, which blasts out of the nozzle. This creates thrust, which pushes the rocket upwards.

Margarine

Margarine *is made of vegetable oils thickened by adding hydrogen.*

Hydrogen peroxide

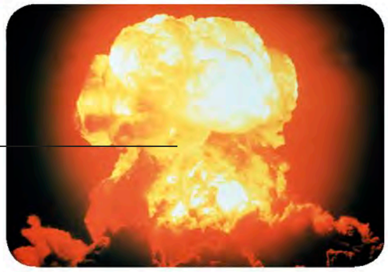
This liquid *is used as a cleaner.*

Delta IV rocket

This powerful rocket *uses 45,460 litres (12,000 gal) of liquid hydrogen as fuel.*

The only **waste product** of hydrogen fuel is **steam**.

This powerful explosion was *created by fusing hydrogen atoms.*



Hydrogen bomb explosion

This energy-efficient bus *runs on a fuel cell fed by hydrogen.*

Hydrogen-powered bus

this element are fused together, releasing heat and light. New stars form inside **nebulae** – such as the **Orion Nebula**. They are clouds of hydrogen gas that slowly collapse in on themselves. Hydrogen gas is the lightest element of all, and much lighter than air. This is why **hydrogen-filled balloons** can fly higher than air-filled ones. Supercold liquid hydrogen is used as **rocket** fuel. Atoms of hydrogen fuse together to produce a lot of energy in **hydrogen bomb** explosions. Pure hydrogen is also a clean energy source used to power some **buses** and cars.